

Developing a VR Simulator for Robotics Navigation and Human Robot Interactions employing Digital Twins

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ABSTRACT

Providing care to seniors and adults with Developmental Disabilities (DD) has seen increased use and development of assistive technologies including service robots. Such robots ease the challenges associated with care, companionship, medication intake, and fall prevention, among others. Research and development in this field rely on in-person data collection to ensure proper robot navigation, interactions, and service. However, the current COVID-19 pandemic has caused the implementation of physical distancing and access restrictions to long-term care facilities, thus making data collection very difficult. This traditional method poses numerous challenges as videos may not be representative of the population in terms of how people move, interact with the environment, or fall. In this paper, we present the development of a VR simulator for robotics navigation and fall detection with digital twins as a solution to test the virtual robot without having access to the real physical location, or real people. The development process required the development of virtual sensors that are able to create LIDAR data for the virtual robot to navigate and detect obstacles. Preliminary testing has allowed us to obtain promising results for the virtual simulator to train a service robot to navigate and detect falls. Our results include virtual maps, robot navigation, and fall detection.

Keywords: Digital twins, Virtual reality, Socially assistive robots, Machine learning.

Index Terms: Computing Methodologies—Computer Graphics—Virtual Reality; Computing Methodologies—Machine Learning

1 INTRODUCTION

Digital twins (DTs) are gaining traction in various industries because of their ability to replicate physical objects and map their features into virtual counterparts [20]. Research including DTs has focused on understanding user acceptance on crisis management practices in Virtual Reality (VR) [15], exploring virtual and real synchronous mapping in manufacturing applications [24], optimization in industrial workstations to design, simulating and optimizing cyber-physical production [11], coaching for physical activity to boost interactivity when exercising [9], and the combination of DTs and healthcare to achieve health management of elderly patients [18], among others.

Providing high-quality care to seniors and adults with Developmental Disabilities (DD) requires personalized care that has been

negatively impacted as a result of the current COVID-19 pandemic and the limited resources many long-term care facilities (LTCFs) face [16]. Emerging technologies such as service robots, VR, and Artificial Intelligence (AI) have been gaining momentum as solutions that can improve the quality of care given the growth of the elderly population. Elderly aged 65 and older account for 16.9% of the Canadian population with a 24% growth projected by 2036 [22].

Research and development associated with service robots have faced a major disruption due to the limited access to LTCFs imposed by COVID-19 restrictions. Such a limitation hinders the evaluation of robot navigation in the environment the robot will be deployed. A solution to this problem is the use of simulation. Tools such as the Robot Operating System (ROS)¹ and Gazebo² allow performing simulations to pilot test and fine-tune the robot's performance. Although 3D simulations are possible with Gazebo, the lack of VR components creates an entry barrier for co-designing safe and friendly environments for people with DD and older adults. Additional limitations include relying on open data sets to train the human-robot interactions (HRI) and fall detection that are not representative of the elderly nor natural falls. As a solution, synthetic data has been gaining momentum as virtual avatars can be modeled to represent various traits of the population along with unique features associated with behaviors that can be predefined with animations, driven by an AI, or teleoperated [2, 23]. Additionally, synthetic data allows overcoming the challenges of collecting data, and the privacy and ethical issues that can limit the sharing of data sets [23].

To overcome the limitations imposed by COVID-19 and to further advance the development of service robots, we build on previous work prototyping a VR simulation environment for virtual service robot detection [8] to expand the functionality to support testing and assessing robot behavior when navigating a virtual environment and detecting a virtual elderly avatar for the purposes of HRI and fall detection. More specifically, we (i) replaced Unity's Sensor SDK with our own sensor implementation to achieve better performance, (ii) created a new virtual environment that replicates a space we currently have access to, and (iii) used the simulator to performing the initial tuning of the robot's navigation stack. Such an approach will enable the use of synthetic data from virtual spaces and avatars.

2 BACKGROUND

Over the past decade, the development of assistive technologies for people with DD, in particular those with cognitive disabilities, and the elderly has drawn significant interest to develop solutions to improve the quality of life [10]. Some of these technologies include the use of social robots, especially those that rely on computer vision for environmental navigation, fall detection, and HRI. Such technologies, however, require data sets with many hundreds to thousands of labeled examples to train high-capacity deep models in a supervised fashion [14]. Usually, data sets are created by capturing live

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¹<https://www.ros.org/>

²<https://gazebo.org/>

data from the different locations where the robot will be deployed. This process ensures the machine learning models will be trained with data that is in a similar domain as the data captured “in the wild”. However, the COVID-19 pandemic has limited access to the LTCFs, thus producing bottlenecks and reliability on publicly available data since capturing information from the location and the users are restricted.

Current trends on DTs have seen the use of synthetic data, immersive technologies, and AI to overcome the limitations of traditional data collection. For example, NVidia’s Omniverse³ is a scalable, multi-GPU real-time reference development platform for 3D simulation and design collaboration. Omniverse provides simulation tools that are being used to assess robotic tasks in VR before deployment to understand and improve shared autonomy between robots and human operators [5]. For example, while in the simulation, operators can observe or intervene allowing researchers to understand the decision-making and outcomes without putting at risk the manufacturing process. Such an approach takes simulation to a new level where immersion and presence are not possible in traditional simulation visualizations without requiring extensive work to produce animations and suitable immersive environments [13].

As a result of the benefits of leveraging synthetic data and simulation with DTs, applications in health care have been gaining momentum. For example, VPepper, a digital replica of the Pepper humanoid robot for elderly care helps train caregivers to avoid excessive wear and tear [4]. Additionally, the use of DTs in patient care can reduce costs as AI-driven solutions can help analyze data from the patients to predict health-related conditions [7], while also increasing compliance with medication intake and health monitoring [18], physical activity [9], and social connectedness and well being [6].

In addition to the current state of DTs and the applications in healthcare aiming to improve the quality of life, off-the-shelf immersive technologies are also making impact. For example, VR headsets with low price points, such as the Oculus Quest 2⁴, and the development of virtual social gathering tools are facilitating the adoption of the technology at a larger scale. Statista reports that in 2021, unit sales of virtual reality (VR) headsets worldwide amount to 6.1 million units, bringing the total cumulative installed base to 16.44 million units [1].

3 MATERIALS AND METHODS

The VR simulation was developed using the Unity game engine (version 2021.1.22f1)⁵. Unlike our previous work where we used Unity’s Sensor SDK to create laser-based sensors such as Time-of-Flight cameras and LIDAR sensors [8], the current version saw the development of our sensors leveraging Unity’s Image Synthesis for Machine Learning⁶. This development decision was made because Unity’s Sensor SDK required powerful GPUs with ray-tracing support, and introduced latency when generating the depth images. Our current implementation is faster and can run in more modest hardware. Additionally, this new approach provides us with control over the simulation without worrying about conflicts between ROS and Unity’s Sensor SDK. There might be incompatibility between libraries, since the Sensor SDK requires a modified version of the HDRP pipeline.

3.1 Aether™, the Service Robot

Aether™ is designed to help seniors and people living with DD to achieve a higher degree of independence, as well as to assist caregivers by alleviating the burden of care. After evaluating the several

³<https://www.nvidia.com/omniverse/>

⁴<https://www.oculus.com/quest-2/>

⁵<https://unity.com/>

⁶<https://bitbucket.org/Unity-Technologies/ml-imagesynthesis/>



(a) Long-term care facility replica



(b) Aether™ replica



(c) Elderly virtual avatar

Figure 1: Environment, service robot, and avatar digital twin representations.

venues where Aether™ can help individuals with DD and their caregivers, together with our community partner, the Developmental Disabilities Association (DDA), we identified two main use-cases: *safety inspection*, where the robot wanders within a facility searching for safety hazards (e.g. objects blocking passage-ways, open doors, and tripping hazards) or fallen people; and *entertainment*, where the robot partakes in cognitive stimulation activities, such as karaoke sessions, trivia games or bingo.

3.2 Digital Twins

Three main DTs were created for the VR simulation including the room to be navigated (a replica of the office space at JDQ), the virtual Aether™, and the elderly virtual avatar. The DTs were created using a combination of Computer Assisted Design software including Blender for the environment, SolidWorks for the service robot, and MakeHuman in combination with Adobe Mixamo for the virtual avatar. Additionally, due to improper texture mapping when importing the 3D assets into Unity, we used a combination of digital image editing tools such as Krita to fix any broken textures. The developed DTs are shown in Fig. 1

With respect to the virtual sensors of the Aether DT, we modeled the LIDAR to produce an error of 2% on the measured distance, as given by its datasheet [21].

3.3 Avatar Movement

Virtual avatar movement is critical to evaluate how the virtual Aether™ recognizes gestures and falls. Two approaches to avatar have been implemented including, i) predefined animations for walking, falling, and standing up were obtained from Adobe’s Mixamo and configured within Unity, and ii) puppeteering employing a VR headset and body motion tracking employing VR controllers mapped to the virtual avatar.

The VR integration was implemented employing the Oculus Integration package with the Unity game engine in conjunction with an Oculus Quest 2. This VR headset was chosen because of its affordability and hand tracking capabilities that will allow us to test HRI interactions in a more realistic manner when puppeteering the virtual avatars. While using the Oculus Quest 2, a user can role-play as

the virtual avatar, see and navigate the environment while observing the virtual Aether™ to the movement. It is worth noting that when using the Oculus Quest 2, the tracking is limited to the head and the hands. Additional tracking can be added employing a combination of HTC Vive Trackers, HTC Vive base stations, and Oculus' Air link streaming features to maintain the wireless experience. The fidelity of the virtual avatar puppeteering will depend on the number of trackers. For example, full-body tracking will require seven trackers for the elbows, waist, knees, and feet.

3.4 ROS Integration and Detection

Aether™ control software architecture was created using ROS Melodic Morenia⁷, and integrated into the simulator employing the ROS TCP Connector package provided by Unity. To guarantee that the simulation does not interfere with the robot software performance, the simulator runs on a Lenovo 81SX000KUS laptop with Microsoft Windows 10, whereas ROS runs on an Intel NUC10i7FNH with Ubuntu 18.04. The communication between both machines is achieved through a direct IEEE 802.3ab (Gigabit Ethernet) connection.

3.5 Preliminary Testing Protocol

To evaluate the simulator regarding its capacity to navigate a virtual environment and detect virtual elderly falls. The virtual service robot will be placed inside a replica of an office space used for real-world testing. The simulation will see the comparison between the maps generated by the real and virtual robots using Cartographer, a system that provides real-time simultaneous localization and mapping (SLAM) in 2D and 3D [12], in addition to identifying parameters for SLAM optimizations, and cost-maps. Additionally, the virtual space will host a virtual elderly avatar who will walk and fall. Finally, gesture recognition will be tested employing the Oculus Quest's hand tracking in front of the virtual service robot.

4 PRELIMINARY RESULTS

4.1 Office Space Map Comparison

Cartographer was set to build a grid map where each cell measured 5 cm². Fig. 2 (a) presents the office space testing environment and Fig. 2 (b) and (c) show the real and virtual maps respectively. The real map has an area of 27.70 m², while the simulated map is 8.2% larger with an area of 29.99 m². This difference is due to some missing obstacles in the simulation and the lack of errors in the simulated LIDAR, discussed in Sect. 3.2.

4.2 Cost-map

We also tested the robot navigation using ROS' default navigation stack (move_base). Figure 2 (d) shows the navigation goal indicated by the red arrow at the top, which is chosen by the user for testing purposes; the global cost-map, where blue and red shades indicate lower and higher navigation costs respectively; the global (i.e. the path the robot should follow to reach the goal) and local plans (i.e. the path the robot will actually follow to achieve the global plan), denoted by green and red lines respectively; and the LIDAR point-cloud represented by the yellow dots. When the user sets a navigation goal, the robot uses the grid map provided by Cartographer to generate the cost map and the global plan. Herein, Dijkstra algorithm is employed for creating the cost map in function of distance and finding the global plan. Once the global plan is found, the robot creates a local plan to guide the robot to the global plan. The global plan is updated at 0.5 Hz, while the local plan is updated at 20 Hz.

⁷<https://wiki.ros.org/melodic>

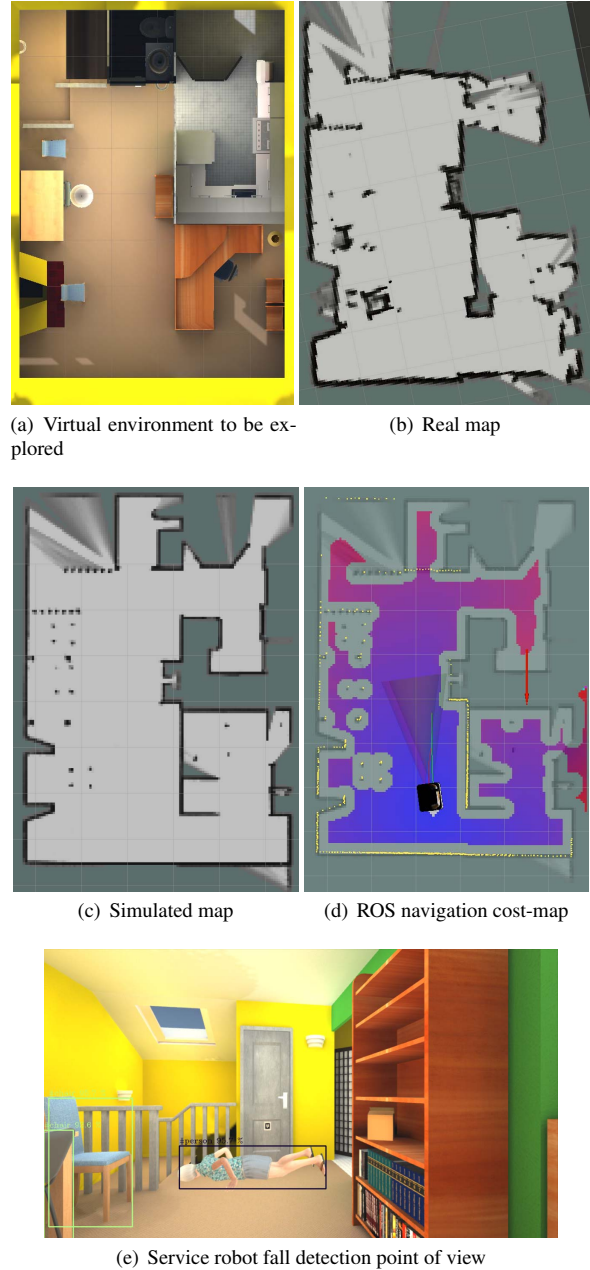


Figure 2: Virtual robot navigation top view and fall detection.

4.3 Fall Detection

To evaluate the person detection, we trained a YOLOv4 [3] object classifier using the COCO [17] and the Fallen Person Data Set (FPDS) [19]. Specifically, we used the official weights provided by YOLOv4 official repository, which is trained on COCO, and then trained the network using FPDS train and test subsets for 7000 epochs. We tested the official and the retrained weights on the FPDS validation subset. As shown in Table 1, retraining classifier using FPDS improved the Average Precision (AP), Precision, and Recall at an IoU (intersection over union) of 0.75 and 0.5.

We used an OpenVino implementation of YOLOv4 and achieved 2 Hz detection rate on CPU. Figure 2 (e) shows that YOLO could

Table 1: Comparison of YOLOv4 classifiers on FPDS validation subset with IoU of 0.75 and 0.5.

Trained on	IoU	AP	Precision	Recall
COCO	0.75	45.3%	58.1%	56.8%
COCO and FPDS	0.75	56.9%	61.8%	68.0%
COCO	0.5	77.2%	81.4%	79.5%
COCO and FPDS	0.5	89.0%	82.6%	90.8%

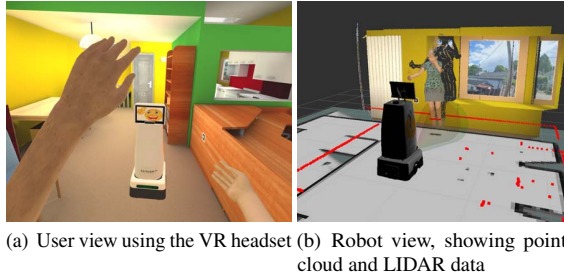


Figure 3: HRI using a VR headset to control the avatar.

successfully detect the fallen person as well as the chairs.

4.4 Gesture Detection

We used the Oculus Integration package to allow the user to interact with the virtual robot by puppeteering an avatar using the VR headset. The gestures performed by the user's hands (e.g. moving the hands, opening and closing fingers) are detected by the controllers, while head gestures (e.g. nodding) are detected by the headset. These gestures are translated into coordinates for the avatar's head and hands, so that it will mimic the user gestures. The VR headset and controllers allow controlling the avatar's head and arms, which increases the realism of the simulation as it allows validating how AetherTM processes and responds to user input that is not based on animations. While using the headset, the user can see through the avatar's eyes, move the avatar's hands using the hand-controllers, and move around the environment using the joystick on the hand controller. Fig. 3 shows the user view on the headset and the robot view.

5 DISCUSSION

Our preliminary results indicate that the DTs adequately recreate the real map, thus allowing for appropriate detection and navigation. We successfully deployed Cartographer to perform SLAM in both real and virtual environments using the same sets of parameters. The robot is also able to build cost maps to explore the environment for possible paths towards a goal. Regarding the fall detection, the use of a virtual avatar walking and falling will speed up the evaluation of Aether fall-detection behavior, since we can simulate several different scenarios that would not be feasible in the real world. Additionally, gesture detection for puppeteering will enable us to test HRI with more dynamic and diverse movements than those provided by predefined animations, thus providing more relevant data to train the robot's behaviors. Finally, we were able to save time and cost associated with real world testing, which includes planning time, setup time, travel time, test execution and test data recording. There is also potential for accelerated and automated test cycles, and automated MTBF (mean time between failures) analysis is promising.

Currently, the simulator faces limitations regarding the accuracy of the sensor models. At this point, we are adopting a Gaussian error model for the LIDAR and RGB-D camera, similar to the model

found on Gazebo. In future works, we will employ an error model that closely resemble the real sensor.

6 CONCLUSION

The use of DTs is gaining momentum in health care applications. In this paper, we present our preliminary results using DTs for understanding how virtual robot navigation, fall detection, and gesture recognition can be tested employing a virtual service robot for elderly care in a VR simulation, as a solution to overcome the COVID-19 restrictions that impede access to LTCFs.

While further investigation is required pertaining to possible optimizations with respect to the SLAM and map accuracy, the integration of the Unity game engine to ROS allowed us to successfully simulate the robot's navigation and virtual avatar detection. We replaced Unity's Sensor SDK with our own implementation of the sensor and were able to improve the performance of the simulator without sacrificing accuracy. The simulator allowed us to tune the Cartographer's parameters in order to improve its SLAM performance. It also allowed us to fine-tune the local planner's parameters, thus allowing AetherTM to improve its obstacle avoidance while producing smoother paths and successfully navigating the environment. We also successfully used YOLOv4 to detect people and objects inside the virtual environment.

Future work will focus on adapting the simulator to produce synthetic images and corresponding annotation to create a data set for improving YOLOv4 detection of fallen people. The images will also introduce new object categories, such as mobility devices (e.g. wheelchairs, canes, walkers) and other objects commonly found in LTCFs. Additionally, we will use other environments and changes to the scene occurring during runtime to test the robot's performance and detection capabilities response. Finally, additional testing will be conducted with full-body tracking to test various human-robot interactions in addition to AI-driven virtual avatars to automate the mobile robot virtual training process.

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